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Storage device and method to check statuses of memory devices

Abstract

A storage device is provided. The storage device includes: a plurality of memory devices configured to output status data indicating a busy state indicating an internal operation is being performed or a ready state; and an interface circuit configured to receive a command for reading the status data of the plurality of memory devices, receive the status data of each of the plurality of memory devices from the plurality of memory devices, and output entire status data indicating statuses of the plurality of memory devices based on the status data of each of the plurality of memory devices.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Korean Patent Application No. 10-2023-0009629, filed in the Korean Intellectual Property Office on Jan. 25, 2023, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) The present disclosure relates to a storage device and an operating method thereof.

2. Description of Related Art

(3) A storage device is a device that stores data under control of a host device such as a computer, a smart phone, and a smart pad. The storage device may include a magnetic disk such as a hard disk drive (HDD), or a semiconductor memory such as a solid state drive (SSD) and a memory card.

(4) In the storage device, a memory controller checks an internal status such as a ready/busy status for each of a plurality of memory devices in order to control the plurality of memory devices in the storage device. Particularly, when one storage device includes a plurality of memory devices, the number of pads corresponding to ready/busy pins of the memory controller increases in order for the memory controller to check the internal status of each of the plurality of memory devices. There is a need for an internal status check processing technology for the plurality of memory devices that may improve an I/O processing speed while reducing the number of the pads of the memory controller.

(5) The memory controller may check a status of each of the plurality of memory devices through a command for checking an internal status. However, the memory controller may check the internal status of the memory device only through a command including a die selection signal for indicating a memory device whose internal status is to be checked among the plurality of memory devices. As the number of memory devices connected to one memory controller increases, overhead required to check internal statuses of a plurality of memory devices increases.

SUMMARY

(6) One or more example embodiments check internal statuses of a plurality of memory devices even without a die selection signal.

(7) One or more example embodiments check internal statuses of a plurality of memory devices even without a separate pin.

(8) One or more example embodiments check internal statuses of a plurality of memory devices connected to a memory controller.

(9) According to an aspect of an example embodiment, a storage device includes: a plurality of memory devices configured to output status data indicating a busy state indicating an internal operation is being performed or a ready state; and an interface circuit configured to receive a command for reading the status data of the plurality of memory devices, receive the status data of each of the plurality of memory devices from the plurality of memory devices, and output entire status data indicating statuses of the plurality of memory devices based on the status data of each of the plurality of memory devices.

(10) According to an aspect of an example embodiment, a storage device includes: a memory controller configured to transmit a first command; and a memory package including: a plurality of memory devices configured to output status data in response to the first command; an interface circuit connected between the plurality of memory devices and the memory controller, and configured to receive the status data from the plurality of memory devices and output a plurality of status data to the memory controller; and a package controller configured to generate a selection signal to control the interface circuit. The status data indicates whether a corresponding memory device is in a ready state or a busy state in which an internal operation is being performed.

(11) According to an aspect of an example embodiment, an operating method of a storage device includes: receiving, by each of a plurality of memory devices, a first command, from a memory controller; outputting, by each of the plurality of memory devices, status data indicating whether a corresponding memory device is in a ready state or a busy state in which an internal operation is being performed, through a status output pin in response to receiving the first command; and generating a selection signal to control a plurality of selectors in an interface circuit to output the status data from each of the plurality of memory devices to the memory controller based on the first command.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The above and other aspects and features will be more apparent from the following description of example embodiments with reference to the attached drawings, in which:
- (2) FIG. 1 illustrates a storage device according to an example embodiment.
- (3) FIG. 2 specifically illustrates the storage device illustrated in FIG. 1 according to an example embodiment.
- (4) FIG. 3 illustrates a memory package according to an example embodiment.
- (5) FIG. 4 illustrates a timing diagram of a chip enable signal, a ready/busy signal of a memory device, and a data signal according to an example embodiment FIG. 3.
- (6) FIG. 5 illustrates a memory package according to an example embodiment.
- (7) FIG. 6 illustrates a timing diagram of a chip enable signal of a memory package, a ready/busy signal of a memory package, and a data signal according to an example embodiment.
- (8) FIG. 7 illustrates a timing diagram of a chip enable signal of a memory package, a ready/busy signal of a memory package, and a data signal according to an example embodiment.
- (9) FIG. 8 illustrates a memory package according to an example embodiment.
- (10) FIG. 9 illustrates a timing diagram of a chip enable signal of a memory package, a ready/busy status of a memory package, and a data signal according to an example embodiment.
- (11) FIG. 10 illustrates a memory package according to an example embodiment.
- (12) FIG. 11 illustrates a timing diagram of a chip enable signal, a ready/busy signal of a memory device, a total ready/busy signal, and a data signal according to an example embodiment.
- (13) FIG. 12 illustrates a block diagram of an SSD system according to an example embodiment.

DETAILED DESCRIPTION

(14) Example embodiments will now be described with reference to the accompanying drawings. Embodiments described herein are example embodiments, and thus, the present disclosure is not limited thereto, and may be realized in various other forms. Each example embodiment provided in the following description is not excluded from being associated with one or more features of another example or another example embodiment also provided herein or not provided herein but consistent with the present disclosure.

(15) Accordingly, the drawings and description are to be regarded as illustrative in nature and not restrictive. Like reference numerals indicate like elements throughout the specification. The term “and/or” includes any and all combinations of one or more of the associated listed items.

Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. For example, the expression, “at least one of a, b, and c,” should be understood as including only a, only b, only c, both a and b, both a and c, both b and c, or all of a, b, and c. In the flowcharts described with reference to the drawings in this specification, the operation order may be changed, various operations may be merged, certain operations may be divided, and certain operations may not be performed.

(16) In addition, a singular form may be intended to include a plural form as well, unless the explicit expression such as “one” or “single” is used. Terms including ordinal numbers such as first, second, and the like will be used only to describe various constituent elements, and are not to be interpreted as limiting these constituent elements. These terms may be used for a purpose of distinguishing one constituent element from other constituent elements.

(17) FIG. 1 illustrates a storage device according to an example embodiment. Referring to FIG. 1, a storage device 1 may store data or process data in response to a command from a host. For example, the storage device 1 may be a solid state drive (SSD), a smart SSD, an embedded multimedia card (eMMC), an embedded universal flash storage (UFS) memory device, a UFS memory card, a compact flash (CF), a secure digital (SD), a micro secure digital (Micro-SD), a

mini secure digital (Mini-SD), an extreme Digital (xD), or a memory stick.

(18) The storage device **1** may include a memory controller **10** and a memory package **20**. The memory controller **10** may control an operation of the storage device **1**. For example, the memory controller **10** may control operations of a plurality of memory devices **40** based on a command, address, and data received from a host device. Specifically, when a command is received from the host device, the memory controller **10** may control each of the plurality of memory devices **40** according to the received command.

(19) The memory controller **10** may perform a status read operation to determine a status of at least one of the plurality of memory devices **40** before or during controlling a specific operation of the plurality of memory devices **40**. For example, the memory controller **10** may transmit a status read command to the plurality of memory devices **40**. In response to the status read command, at least one of the plurality of memory devices **40** may output data indicating a status of the corresponding memory device to the memory controller **10**. In addition, the memory controller **10** may transmit the status read command to the plurality of memory devices **40** to monitor whether an operation is completed.

(20) The memory package **20** may include an interface circuit **30** and a plurality of memory devices (**40_0**, **40_1**, . . . , **40_m**, . . . , **40_n**). For example, one memory package of **20** may include eight memory devices.

(21) The interface circuit **30** may be connected between the plurality of memory devices **40** and the memory controller **10**. The memory controller **10** may be connected to some or all of the plurality of memory devices **40** through the interface circuit **30**.

(22) Each of the plurality of memory devices **40** may store data. In some example embodiments, each of the plurality of memory devices **40** may be a non-volatile memory. In some example embodiments, the non-volatile memory may include a NAND flash memory. In another example embodiment, the non-volatile memory may include an electrically erasable programmable read-only memory (EEPROM), a phase change random access memory (PRAM), a resistive RAM (ReRAM), a resistance random access memory (RRAM), a nano floating gate memory (NFGM), a polymer random access memory (PoRAM), a magnetic random access memory (MRAM), a ferroelectric random access memory (FRAM), or a similar memory.

(23) Each of the plurality of memory devices **40** may perform a write operation, a read operation, or an erase operation under the control of the memory controller **10**. For example, during a write operation, the memory device **40_m** may receive a command, an address, and data from the memory controller **10** and perform the write operation. During a read operation, the memory device **40_m** may receive a command and an address from the memory controller **10**, and may output read data to the memory controller **10**. The memory device **40_m** may be referred to as a chip or die as an individual integrated chip (IC) that has undergone device processing.

(24) In some example embodiments, the memory controller **10** may be disposed in the memory package **20**. In FIG. **1**, it is shown that one memory package **20** is connected to one memory controller **10**, but the present disclosure is not limited thereto, and a plurality of memory packages **20** may be connected to one memory controller **10**.

(25) FIG. **2** illustrates the storage device illustrated in FIG. **1** according to an example embodiment.

(26) Referring to FIG. **2**, the storage device **1** may include the memory controller **10** and the memory package **20**. The memory package **20** may include the interface circuit **30** and the plurality of memory devices **40**. Hereinafter, the memory device **40_0** of the plurality of memory devices **40** will be described as an example. Hereinafter, descriptions of the memory device **400** may be equally applied to each of the plurality of memory devices **40**.

(27) The memory controller **10** may be connected to the interface circuit **30**. The interface circuit **30** may be connected to the plurality of memory devices **40**. The interface circuit **30** may connect a selected one of the plurality of memory devices **40** and the memory controller **10**.

(28) The memory device **40_0** may include a memory interface **41_0**, a control logic **50_0**, and a

memory cell array **60_0**.

(29) The memory controller **10** and the memory device **40_0** may transmit and receive commands CMD, addresses ADDR, and data DATA through a controller interface **11**, the interface circuit **30**, and the memory interface **41_0**.

(30) The memory interface **41_0** may receive a chip enable signal nCE through a first pin **P41_0**. The chip enable signal nCE may be a signal for the memory controller **10** to indicate a specific memory package **20** among a plurality of memory packages connected to the memory controller **10**. Specifically, the chip enable signal nCE may be a signal with which the memory controller **10** instructs the plurality of memory devices **40** in the memory package **20** in order to enable or disable the plurality of memory devices **40** in the specific memory package **20**.

(31) In some example embodiments, the memory interface **41_0** in the memory package **20** selected according to the chip enable signal nCE may transmit and receive signals to and from the memory controller **10** through a second pin **P42_0**, a third pin **P43_0**, a fourth pin **P44_0**, a fifth pin **P45_0**, a sixth pin **P46_0**, a seventh pin **P47_0**, and an eighth pin **P48_0**. For example, the memory interface **41** in the memory package **20** receiving the chip enable signal nCE that is in an enable state (for example, a low level) may transmit/receive signals to/from the memory controller **10** through the second to eighth pins **P42** to **P48**.

(32) The memory interface **41_0** may receive a command latch enable signal CLE through the second pin **P42_0**.

(33) The command latch enable signal CLE may be a signal with which the memory controller **10** instructs the memory device **40_0** to load a command provided to the memory device **40_0** through the seventh pin **P47_0** into a command register of the memory device **40_0**. The command register may store a command received from the memory controller **10**.

(34) The memory interface **41_0** may receive an address latch enable signal ALE through the third pin **P43_0**.

(35) The address latch enable signal ALE may be a signal with which the memory controller **10** instructs the memory device **40_0** to load an address provided to the memory device **40_0** through the seventh pin **P47_0** into an address register. The address register may store a command received from the memory controller **10**.

(36) The memory interface **41_0** may receive a write enable signal nWE through the fourth pin **P44_0**.

(37) The write enable signal nWE may be a signal with which the memory controller **10** instructs the memory device **40_0** so that the command CMD, the address ADDR, and the data DATA are transmitted from the memory controller **10** to the memory device **40_0**.

(38) In some example embodiments, the memory interface **41_0** may obtain the command CMD from a data signal DQ received through the seventh pin **P47_0** in an enable period (for example, a high level state) of the command latch enable signal CLE based on toggle timings of the write enable signal nWE. In addition, the memory interface **41_0** may obtain the address ADDR from a data signal DQ received through the seventh pin **P47_0** in an enable period (for example, a high level state) of the address latch enable signal ALE based on toggle timings of the write enable signal nWE.

(39) The write enable signal nWE may toggle between a high level and a low level, or maintain a static state (for example, a high level or a low level). For example, the write enable signal nWE may toggle during a period in which the command CMD or the address ADDR is transmitted. Accordingly, the memory interface **41_0** may obtain the command CMD or the address ADDR based on the toggle timings of the write enable signal nWE.

(40) The memory interface **41_0** may receive a read enable signal nRE through the fifth pin **P45_0**.

(41) The read enable signal nRE may be a signal with which the memory controller **10** instructs the memory device **40_0** so that data is transmitted from the memory device **40_0** to the memory controller **10**.

- (42) The memory interface **41_0** may receive a data strobe signal nRE through the sixth pin **P46_0**.
- (43) The memory controller **10** or the memory device **40_0** may receive the data signal DQ or transmit the data signal DQ in synchronization with the data strobe signal DQs.
- (44) When the data signal DQ including the data DATA is received from the memory controller **10** in an input operation of the data DATA of the memory device **40_0**, the memory interface **41_0** may receive the data strobe signal DQs that toggles together with the data DATA from the memory controller **10**. The memory interface **41_0** may obtain the data DATA from the data signal DQ based on the toggle timing of the data strobe signal DQs. For example, the memory interface **41_0** may obtain the data DATA by sampling the data signal DQ DQs at rising and falling edges of the data strobe signal DQs.
- (45) In the operation of outputting the data DATA of the memory device **40_0**, when the read enable signal nRE is received from the memory controller **10**, the memory interface **41_0** may generate the data strobe signal DQs that toggles based on the toggle timing of the read enable signal nRE. For example, the memory interface **41_0** may generate the data strobe signal DQs that toggles after a delay of a predetermined period based on a toggling start time of the read enable signal nRE. The memory interface **41_0** may transmit the data signal DQ including the data DATA to the memory controller **10** based on the toggle timing of the data strobe signal DQs. That is, the data DATA may be transmitted to the memory controller **10** in alignment with the toggle timing of the data strobe signal DQs.
- (46) The memory interface **41_0** may receive the data signal nRE through the seventh pin **P47_0**.
- (47) As described above, the data signal DQ may include the command CMD, the address ADDR, and the data DATA. In some example embodiments, the data signal DQ may be transmitted through a plurality of data signal lines.
- (48) In FIG. 2, it is illustrated that the data signal DQ is transmitted and received through one seventh pin **P47_0**, but the present disclosure is not limited thereto, and the data signal DQ may be transmitted and received through a plurality of pins. Hereinafter, pins used to transmit and receive the data signal DQ are referred to as data pins. For example, the memory interface **41_0** may include eight data pins to transmit and receive 8-bit data, and each of a plurality of pins may transmit and receive 1-bit data.
- (49) The memory interface **41_0** may output a ready/busy signal nRNB through an eighth pin **P48_0**. The memory interface **41_0** may transmit status data of the memory device **40_0** to the memory controller **10** through the ready/busy output signal nRNB. In this case, the eighth pin **P48_0** may be referred to as a status output pin of the memory device **40_0**.
- (50) The ready/busy signal nRNB may be a signal indicating the status data of the memory device **40_0**.
- (51) In some example embodiments, when the memory device **40_0** is in a busy state, the memory interface **41_0** may transmit the ready/busy output signal nRNB indicating the busy status to the memory controller **10**. When internal operations of the memory device **40_0** are being performed, the memory device **40_0** may be in a busy state. For example, while the memory device **40_0** reads the data DATA from the memory cell array **60_0** in response to a read command, the memory interface **41_0** may transmit the ready/busy output signal nRNB indicating a busy state (for example, a low level) to the memory controller **10**. For example, while the memory device **40_0** writes the data DATA to the memory cell array **60_0** in response to a write command, the memory interface **41_0** may transmit the ready/busy output signal nRNB indicating a busy state to the memory controller **10**.
- (52) In another example embodiment, when the memory device **40_0** is in a ready state, the memory interface **41_0** may transmit the ready/busy output signal nRNB indicating a ready state to the memory controller **10**. When internal operations of the memory device **40_0** are not performed or have been completed, the memory device **40_0** may be in a ready state. For example, when the memory device **40_0** reads the data DATA from the memory cell array **60_0** for a predetermined

period in response to a read command and the memory interface **41_0** receives the read data **DATA** from the memory cell array **60_0**, the memory interface **41_0** may transmit the ready/busy output signal **nRNB** indicating a ready state (for example, a high level) to the memory controller **10**.

(53) The control logic **50_0** may overall control various operations of the memory device **40_0**. The control logic **50_0** may receive the command **CMD** and the address **ADDR** obtained from the memory interface **41_0**. The control logic **50_0** may generate control signals for controlling other constituent elements inside the memory device **40_0** according to the received command **CMD** and address **ADDR**. For example, the control logic **50_0** may generate various control signals for writing the data **DATA** to each memory cell array **60_0** or reading the data **DATA** from each memory cell array **60_0**.

(54) In some example embodiments, the control logic **50_0** may receive the command **CMD** including a read status command from the memory controller **10**. The control logic **50_0** may generate status data based on the internal status of the memory device **40_0**. The status data may indicate whether the memory device **40_0** is in a busy state in which internal operations are performed or whether the memory device **40_0** is in a ready state in which internal operations are not performed or have been completed.

(55) The memory cell array **60_0** may store the data **DATA** obtained from the memory interface **41_0** under the control of the control logic **50_0**. Each of the memory cell arrays **60_0** may output the stored data **DATA** to the memory interface **41_0** under the control of the control logic **50_0**.

(56) The memory cell array **60_0** may include a plurality of memory cells. For example, the plurality of memory cells may be flash memory cells. However, the present disclosure is not limited thereto, and the memory cells include resistive random access memory (RRAM) cells, ferroelectric random access memory (FRAM) cells, phase change random access memory (PRAM) cells, thyristor random access memory (TRAM) cells, or magnetic random access memory (MRAM) cells. Hereinafter, example embodiments of the present disclosure will be described focusing on an example embodiment in which the memory cells are NAND flash memory cells.

(57) As shown in FIG. 2, channels connected to each of a plurality of first pins **P41** to a plurality of eighth pins **P48** form one channel, and the plurality of memory devices **40** may be connected to the memory controller **10** through the formed channel.

(58) In some example embodiments, when the memory controller **10** transmits a command, data, and/or an address, each of the plurality of memory devices **40** connected to the corresponding channel may receive the command **CMD**, the data **DATA**, and/or the address **ADDR**. Each of the plurality of memory devices **40** may perform a corresponding operation based on the received command **CMD** and address **ADDR**.

(59) The interface circuit **30** may transmit the command, address, and data received from the memory controller **10** to the plurality of memory devices **40** or a selected memory device among the plurality of memory devices **40**, or may transmit the data received from the plurality of memory devices **40** or some memory devices among the plurality of memory devices **40** to the memory controller **10**.

(60) The interface circuit **30** may include a first internal circuit **30-1** to be connected to the memory controller **10**, a second internal circuit **30-2** to be connected to the plurality of memory devices **40**, and a package controller **35**.

(61) The first internal circuit **30-1** may include a first pin **P21** to an eighth pin **P28**. The second internal circuit **30-2** may include a first pin **P31** to an eighth pin **P38**.

(62) The first pin **P31** to the eighth pin **P38** of the interface circuit **30** may correspond to the first pin **P41** to the eighth pin **P48** of the memory interface **41**. The first pin **P31** to the eighth pin **P38** of the interface circuit **30** may be commonly connected to the first pin **P41** to the eighth pin **P48** of each of the plurality of memory devices **40**. The first pin **P21** to the eighth pin **P28** of the interface circuit **30** may correspond to the first pin **P31** to the eighth pin **P38** of the interface circuit **30**.

(63) The package controller **35** is a component for controlling the interface circuit **30**.

(64) In some example embodiments, the package controller **35** may select one of the plurality of memory devices **40** connected to a corresponding channel through a die selection command. When the memory controller **10** transmits the die selection command to the memory package **20**, the interface circuit **30** may generate a selection signal SEL for controlling the interface circuit **30** based on the die selection command. For example, the package controller **35** may control the first internal circuit **30-1** and the second internal circuit **30-2** so that the interface circuit **30** is connected to the memory controller **10** and the plurality of memory devices **40** or the memory controller **10** and some of the plurality of memory devices **40**.

(65) In some example embodiments, the package controller **35** may control the interface circuit **30** based on the command CMD. For example, when a command not including the die selection command is received from the memory controller **10**, the package controller **35** may control the first internal circuit **30-1** and the second internal circuit **30-2** so that it is connected to the memory controller **10** and the plurality of memory devices **40** or the memory controller **10** and some of the plurality of memory devices **40** based on the command.

(66) Then, when the memory controller **10** transmits a command, data, and/or an address, a selected memory device among the plurality of memory devices **40** connected to the corresponding channel may receive the command CMD, the data DATA, and/or the address ADDR.

(67) In FIG. 2, the interface circuit **30** is illustrated as being disposed in the memory package **20**, but the present disclosure is not limited thereto, and the interface circuit **30** may be configured as a separate chip disposed outside the memory package **20**.

(68) As shown in FIG. 2, the memory controller **10** may include the controller interface **11**.

(69) The controller interface **11** may include a first pin P11 to an eighth pin P18. The first pin P11 to the eighth pin P18 of the memory controller **10** may correspond to the first pin P21 to the eighth pin P28 of the interface circuit **30**. Accordingly, the first pin P11 to the eighth pin P18 of the memory controller **10** may correspond to the first pin P41 to the eighth pin P48 of the plurality of memory devices **40**.

(70) The controller interface **11** may transmit the chip enable signal nCE to the plurality of memory devices **40** through the first pin P11. The controller interface **11** may transmit and receive signals through the second pin P12 to the eighth pin P18 of the plurality of memory devices **40** in the memory package **20** selected through the chip enable signal nCE.

(71) The controller interface **11** may respectively transmit the command latch enable signal CLE, the address latch enable signal ALE, and the write enable signal nWE to the plurality of memory devices **40** through the second pin P12, the third pin P13, and the fourth pin P14.

(72) The controller interface **11** may transmit the data signal DQ to the plurality of memory devices **40** or receive the data signal DQ from the plurality of memory devices **40** through the seventh pin P17.

(73) The controller interface **11** may transmit the data signal DQ including the command CMD or the address ADDR to the plurality of memory devices **40** together with the write enable signal nWE that toggles. The controller interface **11** may transmit the data signal DQ including the command CMD to the plurality of memory devices **40** as the command latch enable signal CLE having an enabled status is transmitted, and may transmit the data signal DQ including the address ADDR to the plurality of memory devices **40** as the address latch enable signal ALE having an enabled status is transmitted.

(74) The controller interface **11** may transmit the read enable signal nRE to the plurality of memory devices **40** through the fifth pin P15.

(75) The controller interface **11** may receive the data strobe signal DQs from an arbitrary memory device through the sixth pin P16, or may transmit the data strobe signal DQs to one of the plurality of memory devices **40** or the plurality of memory devices.

(76) In the operation of outputting the data DATA of the memory device **40_0**, the controller interface **11** may generate a toggling read enable signal nRE, and transmit the read enable signal

nRE to the memory device **40_0**. For example, the controller interface **11** may generate the read enable signal nRE that is changed from a fixed state (for example, a high level or low level) to a toggle state before the data DATA is outputted. Accordingly, the data strobe signal DQs that toggles based on the read enable signal nRE may be generated in the memory device **40_0**. The controller interface **11** may receive the data signal DQ including the data DATA together with the data strobe signal DQs that toggles from each of the plurality of memory devices **40**. The controller interface **11** may obtain the data DATA from the data signal DQ based on the toggle timing of the data strobe signal DQs.

(77) In the operation of inputting the data DATA of the memory device **40_0**, the controller interface **11** may generate a toggling data strobe signal DQs. For example, the controller interface **11** may generate the data strobe signal DQs that is changed from a fixed state (for example, a high level or low level) to a toggle state before the data DATA is transmitted. The controller interface **11** may transmit the data signal DQ including the data DATA to the memory device **40_m** based on the toggle timing of the data strobe signal DQs.

(78) The controller interface **11** may receive the ready/busy output signal nRNB from the plurality of memory devices **40** through the eighth pin P18. The controller interface **11** may detect status data of each of the plurality of memory devices **40** based on the ready/busy output signal nRNB.

(79) In some example embodiments, the controller interface **11** may receive status data of each of the plurality of memory devices **40** from each of the plurality of memory devices **40** through the seventh pin P17.

(80) FIG. 3 illustrates a memory package according to an example embodiment.

(81) FIG. 3 illustrates a connection relationship between the interface circuit **30** and the plurality of memory devices **40** in the memory package **20**.

(82) In FIG. 3, it is assumed that one memory package **20** includes eight memory devices (**40_0**, **40_1**, . . . , **40_7**). The present disclosure is not limited thereto, and one memory package **20** may include any number of memory devices.

(83) Each of the plurality of memory devices **40** may include first pins (P41_0, P41_1, . . . , P41_n), a plurality of data pins (P47_0, P47_1, . . . , P47_n), and eighth pins (P48_0, P48_1, . . . , P48_n). FIG. 3 illustrates that one memory device, for example, the memory device **40_0** includes eight data pins (P47a_0, P47b_0, . . . , P47h_0), but the present disclosure is not limited thereto, and each of the plurality of memory devices **40** may include an arbitrary number of data pins.

(84) Hereinafter, in each of the plurality of memory devices **40**, the first data pin is referred to as a first data pin, the second data pin is referred to as a second data pin, the third data pin is referred to as a third data pin, . . . , and the eighth data pin is referred to as an eighth data pin. For example, the first data pin may include a seventh pin P47a_0 of the memory device **40_0**, a seventh pin P47a_1 of the memory device **40_1**, . . . , and a seventh pin P47a_n of the memory device **40_7**. Similarly, the second data pin may include a seventh pin P47b_0 of the memory device **40_0**, a seventh pin P47b_1 of the memory device **40_1**, . . . , and a seventh pin P47b_n of the memory device **40_7**.

(85) As shown in FIG. 3, the interface circuit **30** may include a first selector **401** and a plurality of second selectors (**402_a**, **402_b**, . . . , **402_h**).

(86) As described above, the interface circuit **30** may receive a chip enable signal CE from the memory controller **10** through the first pin P21. The first pin P21 may be connected to the plurality of first pins P41.

(87) The interface circuit **30** may receive the data signal DQ from the memory controller **10** through a plurality of seventh pins P27. Alternatively, the interface circuit **30** may output the data DATA from each of the plurality of memory devices **40** through the plurality of seventh pins P27.

(88) The plurality of memory devices **40** may receive the chip enable signal CE and a plurality of data signals DQ from the interface circuit **30**. The plurality of memory devices **40** may derive the command CMD and/or the address ADDR based on the received data signal DQ.

(89) The interface circuit **30** may output a ready/busy signal received from the plurality of memory

devices **40** through the eighth pin **P28**.

(90) In some example embodiments, the package controller **35** may receive the command **CMD** including a die selection command.

(91) For example, the memory controller **10** may transmit a status read command **CMD** including the die selection command indicating one of the plurality of memory devices **40** to the memory package **20**. The package controller **35** may generate the selection signal **SEL** that controls the interface circuit **30** so that a signal outputted from the memory device selected in response to the die selection command is outputted to the eighth pin **P28**. For example, the package controller **35** may generate the selection signal **SEL** for controlling the first selector **401**.

(92) The first selector **401** may be connected to eighth pins (**P48_0**, **P48_1**, . . . , **P48_n**; **P48**) of each of the plurality of memory devices **40**. The first selector **401** may select one of the outputs received from the plurality of eighth pins **P48** based on the selection signal **SEL** of the package controller **35**, and may output the selected signal to the eighth pin **P28**. In some example embodiments, the first selector **401** may be configured as a multiplexer.

(93) As another example, the memory controller **10** may transmit a read command **CMD** including the die selection command indicating one of the plurality of memory devices **40** to the plurality of memory devices **40**. The package controller **35** may generate the selection signal **SEL** that controls the interface circuit **30** so that a signal outputted from the memory device selected in response to the die selection command is outputted to the seventh pin **P27**. For example, the package controller **35** may generate the selection signal **SEL** for controlling the second selector **402**.

(94) The plurality of second selectors **402** may be connected to data pins of each of the plurality of memory devices **40**. In some example embodiments, data pins (for example, the first data pin, the second data pin, . . . , the eighth data pin, and the like) disposed at the same position in the plurality of memory devices **40** may be connected to each of the plurality of second selectors **402**. In addition, the plurality of second selectors **402** may be connected to each of a plurality of seventh pins (**P27_0**, **P27_1**, . . . , **P27_7**) of the interface circuit **30**. In some example embodiments, each of the plurality of second selectors **402** may be configured as a multiplexer.

(95) For example, the second selector **402_a** may be connected to the first data pin of each of the plurality of memory devices **40**. The second selector **402_a** may select one of the outputs received from the plurality of first data pins based on the selection signal **SEL** of the package controller **35**, and may output the selected signal to the seventh pin **P27_0**.

(96) As another example, the second selector **402_b** may be connected to the second data pin of each of the plurality of memory devices **40**. The second selector **402_b** may select one of the outputs received from the plurality of second data pins based on the selection signal **SEL** of the package controller **35**, and may output the selected signal to the seventh pin **P27_1**.

(97) Similarly, the eighth selector **402_h** may be connected to the eighth data pins. The eighth selector **402_h** may select one of the outputs received from the plurality of eighth data pins based on the selection signal **SEL** of the package controller **35**, and may output the selected signal to the seventh pin **P27_7**.

(98) In some example embodiments, each of the plurality of second selectors **402** may output the data signal **DQ** received through the plurality of seventh pins **P27** based on the selection signal **SEL** of the package controller **35** to the data pin **P47** of the selected memory device.

(99) FIG. 4 illustrates a timing diagram of the chip enable signal, the ready/busy signal of the memory device, and the data signal according to FIG. 3.

(100) Specifically, FIG. 4 illustrates a chip enable signal **CEO**, a data signal **DQ** transmitted to the interface circuit **30** in the memory package **20** selected by the chip enable signal **CEO**, and ready/busy signals (**RNB_0**, **RNB_1**, . . . , **RNB_7**) outputted from each of the plurality of memory devices **40** in the selected memory package **20**. Hereinafter, descriptions of the memory device **40_0** may be equally applied to each of the memory devices **40**.

(101) The ready/busy signal **RNB_0** may indicate the status of the memory device **40_0**. That is,

the ready/busy signal RNB_0 may be a signal outputted from the eighth pin P48_0 of the memory device 40_0. When the ready/busy signal RNB_0 is at a low level, the memory device 400 may indicate a busy state, and when the ready/busy signal RNB_0 is at a high level, the memory device 40_0 may indicate a ready state.

(102) In addition, the memory controller (10 in FIG. 1) may transmit a read command together with the read enable signal (nRE in FIG. 2), and in response to this, the memory interface 41_0 of the memory device 40_0 may transmit the data DATA to the memory controller 10 by aligning with the toggle timing of the data strobe signal DQs based on the read enable signal nRE.

(103) First, at t401, the chip enable signal CEO transitions to an enable level (for example, a low level). The memory controller 10 may select the memory package 20 through the chip enable signal CEO.

(104) While the chip enable signal CEO maintains a low level, the memory controller 10 may transmit the command CMD through the plurality of data signals DQ. The memory controller 10 may transmit a first command 4001 to the memory package 20 through the data signal DQ. In this case, the memory package 20 may be a memory package selected by the chip enable signal CEO. The first command 4001 may include a die selection signal and a read command for selecting die 0. The package controller 35 may generate the selection signal SEL for controlling the first selector 401 and the plurality of second selectors 402 so that the memory controller 10 and the memory device 40_0 are connected based on the die selection signal. The selection signal SEL may be a signal for connecting the plurality of second selectors 402 to the seventh pin P47_0 of the memory device 40_0.

(105) Thereafter, in response to receiving the read command within the first command 4001, the memory device 40_0 may perform an internal operation for preparing data corresponding to the received first command 4001. That is, the memory device 40_0 may be in a busy state. Accordingly, the ready/busy signal RNB_0 indicating the status of the memory device 40_0 may transition to a low level indicating the busy state after a predetermined period after receiving the first command 4001.

(106) At t403, the chip enable signal CEO may transition from a low level to a high level.

(107) At t405, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller 10 may transmit a second command 4003 to the memory package 20 through the data signal DQ. The second command 4003 may include a die selection signal CEBR and a read command for selecting die 1. The package controller 35 may generate the selection signal SEL for connecting the memory controller 10 and the memory device 40_1 in response to the die selection signal. The selection signal SEL may be a signal for connecting the plurality of second selectors 402 to the seventh pin P47_1 of the memory device 40_1.

(108) Thereafter, in response to receiving the read command of the second command 4003, the memory device 40_1 may perform an internal operation for preparing data corresponding to the received second command 4003. Accordingly, the ready/busy signal RNB_1 indicating the status of the memory device 40_1 may transition to a low level after a predetermined period after receiving the second command 4003.

(109) At t407, the chip enable signal CEO may transition from a low level to a high level.

(110) At t409, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller 10 may transmit a third command 4005 to the memory package 20 through the data signal DQ. The third command 4005 may include a die selection signal CEBR and a read command for selecting die 7. The package controller 35 may generate the selection signal SEL for connecting the memory controller 10 and the memory device 40_7 based on the die selection signal. The selection signal SEL may be a signal for connecting the plurality of second selectors 402 to the seventh pin P47_7 of the memory device 40_7.

(111) In response to receiving the read command of the third command **4005**, the memory device **40_7** may perform an internal operation for preparing data corresponding to the received third command **4005**. Accordingly, the ready/busy signal **RNB_7** indicating the status of the memory device **407** may transition to a low level after a predetermined period after receiving the third command **4005**.

(112) At **t411**, the chip enable signal **CEO** may transition from a low level to a high level.

(113) At **t413**, the chip enable signal **CEO** may transition from a high level to a low level. While the chip enable signal **CEO** is maintained at the low level, the memory controller **10** may transmit a fourth command **4007** to the memory package **20** through the data signal **DQ**. The fourth command **4007** may include a die selection signal **CEBR** and a status read command **70h** for selecting die **1**. The package controller **35** may generate the selection signal **SEL** for connecting the memory controller **10** and the memory device **40_1** based on the die selection signal. The selection signal **SEL** may be a signal for connecting the first selector **401** to the eighth pin **P48_1** of the memory device **40_1**. In some example embodiments, the memory device **40_1** may output status data **4009** indicating the status of the memory device **40_1** at a time point of receiving the status read command **70h** or at any time point (for example, **t414**) while receiving the status read command **70h**. The status data **4009** may indicate a busy state, which is the status of the memory device **40_1** at **t414**.

(114) At **t415**, the chip enable signal **CEO** may transition from a low level to a high level.

(115) At **t417**, the chip enable signal **CEO** may transition from a high level to a low level. While the chip enable signal **CEO** is maintained at the low level, the memory controller **10** may transmit a fifth command **4011** to the memory package **20** through the data signal **DQ**. The fifth command **4011** may include a die selection signal **CEBR** and a status read command **70h** for selecting die **0**. The package controller **35** may generate the selection signal **SEL** for connecting the memory controller **10** and the memory device **40_0** based on the die selection signal. The selection signal **SEL** may be a signal for connecting the first selector **401** to the eighth pin **P48_0** of the memory device **40_0**. In some example embodiments, the memory device **400** may output status data **4013** indicating the status of the memory device **40_0** at any time point (for example, **t418**) while receiving the status read command **70h**. The status data **4013** may indicate a ready state, which is the status of the memory device **40_0** at **t418**.

(116) Then, at **t419**, the chip enable signal **CEO** may transition from a high level to a low level. While the chip enable signal **CEO** is maintained at the low level, the memory controller **10** may transmit a sixth command **4015** to the memory package **20** through the data signal **DQ**. The sixth command **4015** may include a die selection signal **CEBR** for selecting die **0**. In response to the sixth command **4015**, the package controller **35** may generate the selection signal **SEL** for connecting the memory controller **10** and the memory device **40_0** based on the die selection signal. The selection signal **SEL** may be a signal for controlling the plurality of second selectors **402** so as to output the data outputted from the memory device **40_0** having performed the read operation in response to the sixth command **4015** as the data signal **DQ** through the plurality of seventh pins **P27**.

(117) FIG. 5 illustrates a memory package according to an example embodiment.

(118) FIG. 5 illustrates a connection relationship between the interface circuit **31** and the plurality of memory devices **40** in the memory package **100**.

(119) In FIG. 5, it is assumed that one memory package **100** includes eight memory devices (**40_0**, **40_1**, . . . , **40_7**). The present disclosure is not limited thereto, and one memory package **100** may include any number of memory devices.

(120) Each of the plurality of memory devices **40** may include first pins (**P41_0**, **P41_1**, . . . , **P41_n**), a plurality of data pins (**P47_0**, **P47_1**, . . . , **P47_n**), eighth pins (**P48_0**, **P48_1**, . . . , **P48_n**), and ninth pins (**P49_0**, **P49_1**, . . . , **P49_n**). FIG. 5 illustrates that one memory device, for example, the memory device **40_0** includes eight data pins (**P47a_0**, **P47b_0**, . . . , **P47h_0**), but the

present disclosure is not limited thereto, and each of the plurality of memory devices **40** may include an arbitrary number of data pins.

(121) Hereinafter, descriptions of the memory device **40_0** may be equally applied to each of the memory devices **40**.

(122) The memory device **40_0** may output an error signal nERR through the ninth pin **P49_0**. The memory device **40_0** may transmit error data of the memory device **40_0** to the memory controller **10** through the error signal nERR. The error data may include information about a component in which an error occurred among components in the memory device **40_0**. The error may occur when an arbitrary sequence is broken while processing received data when the memory device **40_0** writes data to the memory cell array **60_0** in response to a write command.

(123) Hereinafter, in each of the plurality of memory devices **40**, the first data pin is referred to as a first data pin, the second data pin is referred to as a second data pin, the third data pin is referred to as a third data pin, . . . , and the eighth data pin is referred to as an eighth data pin. For example, the first data pin may include a seventh pin **P47a_0** of the memory device **40_0**, a seventh pin **P47a_1** of the memory device **40_1**, . . . , and a seventh pin **P47a_n** of the memory device **40_7**. Similarly, the second data pin may include a seventh pin **P47b_0** of the memory device **400**, a seventh pin **P47b_1** of the memory device **40_1**, . . . , and a seventh pin **P47b_n** of the memory device **40_7**.

(124) As shown in FIG. 5, the interface circuit **31** may include a plurality of second selectors (**402_a**, **402_b**, . . . , **402_h**), a third selector **403**, a fourth selector **405**, and a fifth selector **407**.

(125) The interface circuit **31** may receive the chip enable signal CE from the memory controller **10** through the first pin **P21**. The first pin **P21** may be connected to a plurality of first pins **P41** of the plurality of memory devices **40**.

(126) The interface circuit **31** may receive the data signal DQ from the memory controller (**10** in FIG. 3) through a plurality of seventh pins **P27**. Alternatively, the interface circuit **31** may output the data DATA from each of the plurality of memory devices **40** through the plurality of seventh pins **P27**.

(127) The plurality of memory devices **40** may receive the chip enable signal CE and a plurality of data signals DQ from the interface circuit **31**. The plurality of memory devices **40** may derive the command CMD and/or the address ADDR based on the received data signal DQ.

(128) The interface circuit **31** may output a ready/busy signal RNB or an error signal ERR received from each of the plurality of memory devices **40** through the plurality of seventh pins **P27**.

(129) In some example embodiments, the plurality of memory devices **40** may receive a command CMD that does not include a separate die selection command CEBR.

(130) For example, the memory controller **10** may transmit an entire status read command CMD to each of the plurality of memory devices **40**. The package controller **35** may generate a selection signal SEL2 that controls the third selector **403** so that signals outputted from the eighth pin **P48** are outputted to the plurality of seventh pins **P27**. For example, the package controller **35** may generate the second selection signal SEL2 that controls the third selector **403** to be connected to the fourth selector **405**.

(131) In another example embodiment, the memory controller **10** may transmit an entire error read command CMD to each of the plurality of memory devices **40**. The package controller **35** may generate a selection signal SEL2 that controls the third selector **403** so that signals outputted from the ninth pin **P49** are outputted to the plurality of seventh pins **P27**. For example, the package controller **35** may generate a second selection signal SEL2 that controls the third selector **403** to be connected to the fifth selector **407**.

(132) The third selector **403** may be connected to the fourth selector **405** and the fifth selector **407**. The third selector **403** may be connected to the plurality of seventh pins **P27**. The third selector **403** may connect the plurality of seventh pins **P27** to an output terminal of the fourth selector **405** or an output terminal of the fifth selector **407** based on the second selection signal SEL2. In some example embodiments, the third selector **403**, the fourth selector **405**, and the fifth selector **407**

may be configured as multiplexers.

(133) In some example embodiments, the fourth selector **405** may be connected to the eighth pins (**P48_0**, **P48_1**, . . . , **P48_n**) of each of the plurality of memory devices **40**. For example, because memory package **100** includes eight memory devices (**40_0**, **40_1**, . . . , **40_7**), the fourth selector **405** may be connected to eight eighth pins (**P48_0**, **P48_1**, . . . , **P48_7**). Accordingly, status data of each of the eight memory devices (**40_0**, **40_1**, . . . , **40_7**) may be input to the fourth selector **405**.

(134) In some example embodiments, the fifth selector **407** may be connected to the ninth pin **P49** of each of the plurality of memory devices **40**. For example, because the memory package **100** includes the eight memory devices (**40_0**, **40_1**, . . . , **40_7**), the fifth selector **407** may be connected to the eight ninth pins (**P49_0**, **P49_1**, . . . , **P49_7**). Accordingly, error data of each of the eight memory devices (**40_0**, **40_1**, . . . , **40_7**) may be input to the fifth selector **407**.

(135) The plurality of second selectors **402** may be connected to data pins of each of the plurality of memory devices **40**. In addition, the plurality of second selectors **402** may be connected to each of the plurality of seventh pins (**P27_0**, **P27_1**, . . . , **P27_7**) of the interface circuit **30**. The package controller **35** may generate a first selection signal **SEL1** that controls the plurality of second selectors **402** so that a signal outputted from the memory device is selected and provided to the plurality of seventh pins **P27**.

(136) FIG. **6** illustrates a timing diagram of a chip enable signal of a memory package, a ready/busy signal of a memory package, and a data signal according to FIG. **5**.

(137) Specifically, FIG. **6** illustrates a chip enable signal **CEO**, a data signal **DQ** transmitted to the interface circuit **31** in the memory package **100** selected by the chip enable signal **CEO**, and ready/busy signals (**RNB_0**, **RNB_1**, . . . , **RNB_7**) outputted from each of the plurality of memory devices **40** in the selected memory package **100**. Hereinafter, descriptions of the memory device **40_0** may be equally applied to each of the memory devices **40**.

(138) In addition, the memory controller (**10** in FIG. **1**) may transmit a read command together with the read enable signal (**nRE** in FIG. **2**), and in response to this, the memory interface **41_0** of the memory device **40_0** may transmit the data **DATA** to the memory controller **10** by aligning with the toggle timing of the data strobe signal **DQs** based on the read enable signal **nRE**.

(139) First, at **t601**, the chip enable signal **CEO** transitions to an enable level (for example, a low level). The memory controller **10** may select the memory package **100** through the chip enable signal **CEO**.

(140) While the chip enable signal **CEO** maintains a low level, the memory controller **10** may transmit the command **CMD** through the plurality of data signals **DQ**. The memory controller **10** may transmit a first command **6001** to the memory package **100** through the data signal **DQ**. In this case, the memory package **100** may be a memory package selected by the chip enable signal **CEO**. The first command **6001** may include a die selection signal **CEBR** and a read command for selecting die **0**. The package controller **35** may generate a first selection signal **SEL1** for connecting the memory controller **10** and the memory device **40_0** based on the die selection signal **CEBR**. The first selection signal **SEL1** may be a signal for connecting the plurality of second selectors **402** to the seventh pin **P47_0** of the memory device **40_0**.

(141) Thereafter, in response to receiving the read command within the first command **6001**, the memory device **40_0** may perform an internal operation for preparing data corresponding to the received first command **6001**. That is, the memory device **40_0** may be in a busy state.

Accordingly, the ready/busy signal **RNB_0** indicating the status of the memory device **40_0** may transition to a low level indicating the busy state after a predetermined period after receiving the first command **6001**.

(142) At **t603**, the chip enable signal **CEO** may transition from a low level to a high level.

(143) At **t605**, the chip enable signal **CEO** may transition from a high level to a low level. While the chip enable signal **CEO** is maintained at the low level, the memory controller **10** may transmit a second command **6003** to the memory package **100** through the data signal **DQ**. The second

command **6003** may include a die selection signal CEBR and read command for selecting die **1**. The package controller **35** may generate a first selection signal SEL1 for connecting the memory controller **10** and the memory device **40_1** based on the die selection signal CEBR. The first selection signal SEL1 may be a signal for connecting the plurality of second selectors **402** to the seventh pin P47_1 of the memory device **40_1**.

(144) Thereafter, in response to receiving the read command within the second command **6003**, the memory device **40_1** may perform an internal operation for preparing data corresponding to the received second command **6003**. Accordingly, the ready/busy signal RNB_1 indicating the status of the memory device **40_1** may transition to a low level after a predetermined period after receiving the second command **6003**.

(145) At t607, the chip enable signal CEO may transition from a low level to a high level.

(146) At t609, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a third command **6005** to the memory package **100** through the data signal DQ. The third command **6005** may include a die selection signal CEBR and a read command for selecting die **7**. The package controller **35** may generate a first selection signal SEL1 for connecting the memory controller **10** and the memory device **40_7** based on the die selection signal CEBR. The first selection signal SEL1 may be a signal for connecting the plurality of second selectors **402** to the seventh pin P47_7 of the memory device **40_7**.

(147) In response to receiving the read command of the third command **6005**, the memory device **40_7** may perform an internal operation for preparing data corresponding to the received third command **6005**. Accordingly, the ready/busy signal RNB_7 indicating the status of the memory device **407** may transition to a low level after a predetermined period after receiving the third command **6005**.

(148) At t611, the chip enable signal CEO may transition from a low level to a high level.

(149) At t613, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a fourth command **6007** to the memory package **100** through the data signal DQ. The fourth command **6007** may include an entire status read command **90h**. The package controller **35** may generate a second selection signal SEL2 for connecting the memory controller **10** and the plurality of memory devices **40**. For example, the second selection signal SEL2 may be a signal for connecting the third selector **403** to the fourth selector **405**.

(150) In some example embodiments, the memory device **400** may output status data indicating the status of the memory device **40_0** at a time point of receiving the entire status read command **90h** or at any time point (for example, t614) while receiving the entire status read command **90h**.

(151) In some example embodiments, the plurality of memory devices **40** may output entire status data **6009** in response to receiving the fourth command **6007**. The entire status data **6009** may indicate the status of each of the plurality of memory devices **40** at t614. Specifically, because the memory device **40_0** is in a ready state, the seventh pin P27_0 may output a bit (for example, 1) indicating the ready state. Because the memory devices (**40_1**, . . . , **40_7**) at t614 are in a busy state, the seventh pin (P27_1, . . . , P27_7) may output a bit (for example, 0) indicating the busy state.

(152) The memory controller **10** may detect status data of each of the memory devices **40** based on the entire status data **6009** outputted from the memory package **100** in response to the fourth command **6007**. The memory controller **10** does not obtain status data indicating the status of one memory device **40_0**, but may obtain entire status data indicating the status of each of the plurality of memory devices **40** through the entire status read command. For example, the memory controller **10** may detect the status data of the memory device **40_0** through data outputted through the seventh pin P27_0, may detect the status data of the memory device **40_1** through data outputted through the seventh pin P27_1, and may detect the status data of the memory device **40_2** through

data outputted through the seventh pin P27_3.

(153) Subsequently, at t615, the chip enable signal CEO may transition from a low level to a high level.

(154) At t617, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller 10 may transmit a fifth command 6011 to the memory package 100 through the data signal DQ. The fifth command 6011 may include an entire status read command 90h. The package controller 35 may generate a second selection signal SEL2 for connecting the memory controller 10 and the plurality of memory devices 40. For example, the second selection signal SEL2 may be a signal for connecting the third selector 403 to the fourth selector 405.

(155) In some example embodiments, the plurality of memory devices 40 may output entire status data 6013 in response to receiving the fifth command 6011. The entire status data 6013 may indicate the status of each of the plurality of memory devices 40 at t618. Specifically, because the memory devices 40_0 and 40_1 are in a ready state, the seventh pins P27_0 and P27_1 may output a bit (for example, 1) indicating the ready state. Because the memory devices (40_2, . . . , 40_7) at t614 are in a busy state, the seventh pin (P27_2, . . . , P27_7) may output a bit (for example, 0) indicating the busy state.

(156) Then, at t619, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller 10 may transmit a sixth command 6015 to the memory package 100 through the data signal DQ. The sixth command 6015 may include a die selection signal CEBR for selecting die 0. In response to the sixth command 6015, the package controller 35 may generate a first selection signal SEL1 for connecting the memory controller 10 and the memory device 40_0. For example, the first selection signal SEL1 may be a signal for connecting the plurality of second selectors 402 to data pins of the memory device 40_0.

(157) Then, the memory device 40_0 may output data 6017 corresponding to the first command 6001 through the seventh pin P47_0. The interface circuit 31 may output data outputted from the memory device 40_0 as the data 6017 through the plurality of second selectors 402 and the plurality of seventh pin P27.

(158) FIG. 7 illustrates a timing diagram of a chip enable signal of a memory package, a ready/busy signal of a memory package, and a data signal according to FIG. 5.

(159) Specifically, FIG. 7 illustrates a chip enable signal CEO, a data signal DQ transmitted to the interface circuit 30 in the memory package 100 selected by the chip enable signal CEO, and ready/busy signals (RNB_0, RNB_1, . . . , RNB_7) outputted from each of the plurality of memory devices 40 in the selected memory package 100. Hereinafter, descriptions of the memory device 40_0 may be equally applied to each of the memory devices 40.

(160) In addition, the memory controller 10 may transmit a write command together with the write enable signal (nWE of FIG. 2), and in response to this, the memory interface 41_0 of the memory device 40_0 may receive data DATA in alignment with the toggle timing of the write enable signal nRE. Then, each of the plurality of memory devices 40 may perform a write operation.

(161) First, at t701, the chip enable signal CEO transitions to an enable level (for example, a low level). The memory controller 10 may select the memory package 100 through the chip enable signal CEO.

(162) While the chip enable signal CEO is maintained at the low level, the memory controller 10 may transmit the command CMD through the plurality of data signals DQ. The memory controller 10 may transmit a first command 7001 to the memory package 100 through the data signal DQ. In this case, the memory package 100 may be a memory package selected by the chip enable signal CEO. The first command 7001 may include a die selection signal CEBR and a write command for selecting die 0. The package controller 35 may generate a first selection signal SEL1 for connecting the memory controller 10 and the memory device 40_0 based on the die selection signal CEBR.

The first selection signal SEL1 may be a signal for connecting the plurality of second selectors 402 to the seventh pin P47_0 of the memory device 40_0.

(163) In response to receiving the write command within the first command 7001, the memory device 40_0 may perform an internal operation for writing data corresponding to the received first command 7001. That is, the memory device 40_0 may be in a busy state. Accordingly, the ready/busy signal RNB_0 indicating the status of the memory device 40_0 may transition to a low level indicating the busy state after a predetermined period after receiving the first command 7001.

(164) At t703, the chip enable signal CEO may transition from a low level to a high level.

(165) At t705, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller 10 may transmit a second command 7003 to the memory package 100 through the data signal DQ. The second command 7003 may include a die selection signal CEBR and a write command for selecting die 1. The package controller 35 may generate a first selection signal SEL1 for connecting the memory controller 10 and the memory device 40_1 based on the die selection signal CEBR. The first selection signal SEL1 may be a signal for connecting the plurality of second selectors 402 to the seventh pin P47_1 of the memory device 40_1.

(166) Thereafter, in response to receiving the write command within the second command 7003, the memory device 40_1 may perform an internal operation for writing data corresponding to the received second command 7003. Accordingly, the ready/busy signal RNB_1 indicating the status of the memory device 40_1 may transition to a low level after a predetermined period after receiving the second command 7003.

(167) At t707, the chip enable signal CEO may transition from a low level to a high level.

(168) At t709, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller 10 may transmit a third command 7005 to the memory package 100 through the data signal DQ. The third command 7005 may include a die selection signal CEBR and a write command for selecting die 7. The package controller 35 may generate a first selection signal SEL1 for connecting the memory controller 10 and the memory device 40_7 based on the die selection signal CEBR. The first selection signal SEL1 may be a signal for connecting the plurality of second selectors 402 to the seventh pin P47_7 of the memory device 40_7.

(169) In response to receiving the write command within the third command 7005, the memory device 40_7 may perform an internal operation for writing data corresponding to the received third command 7005. Accordingly, the ready/busy signal RNB_7 indicating the status of the memory device 40_7 may transition to a low level after a predetermined period after receiving the third command 7005.

(170) At t711, the chip enable signal CEO may transition from a low level to a high level.

(171) At t713, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller 10 may transmit a fourth command 7007 to the memory package 100 through the data signal DQ. The fourth command 7007 may include an entire error read command 98h. The package controller 35 may generate a second selection signal SEL2 for connecting the memory controller 10 and the plurality of memory devices 40. For example, the second selection signal SEL2 may be a signal for connecting the third selector 403 to the fifth selector 407.

(172) In some example embodiments, the memory device 400 may output error data including error information of the memory device 40_0 at a time point of receiving the entire error read command 98h or at any time point (for example, t714) while receiving the entire error read command 98h.

(173) In some example embodiments, the plurality of memory devices 40 may output entire error data 7009 in response to receiving the fourth command 7007. The entire error data 7009 may indicate whether an error exists in each of the plurality of memory devices 40 at t714. Specifically, when an error exists in the memory device 40_0, the seventh pin P27_0 may output a bit (for

example, 1) indicating that there is an error. When no error exists in the memory devices (**40_1**, . . . , **40_7**), the seventh pin (**P27_1**, . . . , **P27_7**) may output a bit (for example, 0) indicating no error. The memory controller **10** does not obtain status data indicating the error of one memory device **400**, but may obtain entire error data indicating the status of each of the plurality of memory devices **40** through the entire error read command. For example, the memory controller **10** may detect the error data of the memory device **40_0** through data outputted through the seventh pin **P27_0**, may detect the error data of the memory device **40_1** through data outputted through the seventh pin **P27_1**, and may detect the error data of the memory device **40_2** through data outputted through the seventh pin **P27_2**.

(174) The memory controller **10** may detect error data of each of the memory devices **40** based on the entire error data **7009** outputted from the memory package **100** in response to the fourth command **7007**.

(175) Subsequently, at **t715**, the chip enable signal **CEO** may transition from a low level to a high level.

(176) At **t717**, the chip enable signal **CEO** may transition from a high level to a low level. While the chip enable signal **CEO** is maintained at the low level, the memory controller **10** may transmit a fifth command **7011** to the memory package **100** through the data signal **DQ**. The fifth command **7011** may include an entire error read command **98h**. The package controller **35** may generate a second selection signal **SEL2** for connecting the memory controller **10** and the plurality of memory devices **40**. For example, the second selection signal **SEL2** may be a signal for connecting the third selector **403** to the fifth selector **407**.

(177) In some example embodiments, the plurality of memory devices **40** may output error data **7013** in response to receiving the fifth command **7011**. The error data **7013** may indicate whether an error exists in each of the plurality of memory devices **40** at **t718**.

(178) FIG. **8** illustrates a memory package according to an example embodiment.

(179) FIG. **8** illustrates a connection relationship between an interface circuit **32** and a plurality of memory devices **40** in a memory package **200**.

(180) In FIG. **8**, it is assumed that one memory package **200** includes eight memory devices (**40_0**, **40_1**, . . . , **40_7**). The present disclosure is not limited thereto, and one memory package **200** may include any number of memory devices.

(181) Each of the plurality of memory devices **40** may include first pins (**P41_0**, **P41_1**, . . . , **P41_n**) and a plurality of data pins (**P47_0**, **P47_1**, . . . , **P47_n**). FIG. **8** illustrates that one memory device, for example, the memory device **40_0**, includes eight data pins (**P47a_0**, **P47b_0**, . . . , **P47h_0**), but the present disclosure is not limited thereto, and each of the plurality of memory devices **40** may include an arbitrary number of data pins. Because the plurality of memory devices **40** in the memory package **200** do not include a pin indicating status data of each memory device, a separate command **CMD** may be required to detect status data of each of the plurality of memory devices **40**.

(182) Hereinafter, descriptions of the memory device **40_0** may be equally applied to each of the memory devices **40**.

(183) In some example embodiments, the memory controller **10** may transmit a status read command to the plurality of memory devices **40**. In response to the status read command, at least one of the plurality of memory devices **40** may output status data of the corresponding memory device to the memory controller **10**. In another example embodiment, the memory controller **10** may transmit an error read command to the plurality of memory devices **40**. In response to the error read command, at least one of the plurality of memory devices **40** may output error data of the corresponding memory device to the memory controller **10**.

(184) For example, the memory device **400** may output status data of the memory device **40_0** through the seventh pin **P47a_0** and error data of the memory device **400** through the seventh pin **P47g_0**. However, the present disclosure is not limited thereto, and the memory device **40_0** may

output status data and error data through an arbitrary data pin.

(185) As shown in FIG. 8, the interface circuit 32 may include a sixth selector 409.

(186) The interface circuit 32 may receive the chip enable signal CE from the memory controller 10 through the first pin P21. The first pin P21 may be connected to a plurality of first pins (P41_0, P41_1, . . . , P41_n; P41) of the plurality of memory devices 40.

(187) The interface circuit 32 may receive the data signal DQ from the memory controller (10 in FIG. 1) through a plurality of seventh pins (P27_0, P27_1, . . . , P27_7; P27). Alternatively, the interface circuit 30 may output data DATA from each of the plurality of memory devices 40 through the plurality of seventh pins P27.

(188) The plurality of memory devices 40 may receive the chip enable signal CE and the plurality of data signals DQ from the interface circuit 32. The plurality of memory devices 40 may derive the command CMD and/or the address ADDR based on the received data signal DQ. The package controller 35 may generate a selection signal SEL for controlling the sixth selector 409 based on the command CMD.

(189) The interface circuit 30 may output status data and/or error data of the plurality of memory devices 40 through the plurality of seventh pins P27.

(190) The sixth selector 409 may select eight signals outputted from the plurality of data pins (P47a_0, P47b_0, . . . , P47h_0; P47a_1, P47b_1, . . . , P47b_h; P47a_n, P47b_n, . . . , P47h_n) based on the selection signal SEL, and may respectively connect the selected signals to the plurality of seventh pins P27. In some example embodiments, the sixth selector 409 may include a number of input pins corresponding to the number of output pins of the plurality of memory devices 40 disposed within one memory package 200. In some example embodiments, the sixth selector 409 may be configured as a multiplexer.

(191) In some example embodiments, the package controller 35 may generate a selection signal SEL based on corresponding information between pins of the plurality of memory devices 40 and input pins of the sixth selector 409. For example, the package controller 35 may include a memory storing corresponding information about which output of a plurality of outputs of the plurality of memory devices 40 is inputted to each input pin of the sixth selector 409. The sixth selector 409 may select and output as many signals as the number of output pins (that is, the seventh pins (P27_0, P27_1, . . . , P27_7)) among the signals inputted through the input pins based on the selection signal SEL. The sixth selector 409 may output the selected output signal to the memory controller 10 through the seventh pin P27.

(192) In some example embodiments, the plurality of memory devices 40 may receive a command CMD that does not include a separate die selection command CEBR.

(193) For example, the memory controller 10 may transmit an entire status read command CMD to each of the plurality of memory devices 40. A control logic 52 may control the sixth selector 409 so that signals outputted from the first data pins (P47a_0, P47a_1, . . . , P47a_n) of each of the plurality of memory devices 40 are outputted to the plurality of seventh pins P27.

(194) As another example, the memory controller 10 may transmit the entire error read command CMD to each of the plurality of memory devices 40. The control logic 52 may control the sixth selector 409 so that signals outputted from the seventh data pins (P47g_0, P47g_1, . . . , P47g_n) of each of the plurality of memory devices 40 are outputted to the plurality of seventh pins P27.

(195) FIG. 9 illustrates a timing diagram of a chip enable signal of a memory package, a ready/busy status of a memory package, and a data signal according to FIG. 8.

(196) Specifically, FIG. 9 illustrates a chip enable signal CEO, a data signal DQ transmitted to the interface circuit 30 in the memory package 200 selected by the chip enable signal CEO, and a ready/busy state of each of the plurality of memory devices 40 in the selected memory package 200. Hereinafter, descriptions of the memory device 40_0 may be equally applied to each of the memory devices 40.

(197) In addition, the memory controller 10 may transmit a read command together with the read

enable signal (nRE in FIG. 2), and in response to this, the memory interface **41_0** of the memory device **400** may transmit the data DATA to the memory controller **10** by aligning with the toggle timing of the data strobe signal DQs based on the read enable signal nRE.

(198) First, at **t901**, the chip enable signal CEO transitions to an enable level (for example, a low level). The memory controller **10** may select the memory package **100** through the chip enable signal CEO.

(199) While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit the command CMD through the plurality of data signals DQ. The memory controller **10** may transmit a first command **9001** to the memory package **200** through the data signal DQ. In this case, the memory package **200** may be a memory package selected by the chip enable signal CEO. The first command **9001** may include a die selection signal CEBR and a read command for selecting die **0**. The package controller **35** may generate a selection signal SEL for controlling the sixth selector **409** to output an input of the seventh pin **P47** of the memory device **40_0** based on the command CMD and corresponding information.

(200) Thereafter, in response to receiving the read command within the first command **9001**, the memory device **40_0** may perform an internal operation for preparing data corresponding to the received first command **9001**. That is, the memory device **40_0** may be in a busy state.

(201) At **t903**, the chip enable signal CEO may transition from a low level to a high level.

(202) At **t905**, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a second command **9003** to the memory package **200** through the data signal DQ. The second command **9003** may include a die selection signal CEBR and read command for selecting die **1**. The package controller **35** may generate a selection signal SEL for controlling the sixth selector **409** to receive and output an input of the seventh pin **P47** of the memory device **40_1** based on the die selection signal CEBR and corresponding information.

(203) In response to receiving the read command within the second command **9003**, the memory device **40_1** may perform an internal operation for preparing data corresponding to the received second command **9003**. Accordingly, the memory device **40_1** may be in a busy state.

(204) At **t907**, the chip enable signal CEO may transition from a low level to a high level.

(205) At **t909**, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a third command **9005** to the memory package **200** through the data signal DQ. The third command **9005** may include a die selection signal CEBR and a read command for selecting die **7**. The package controller **35** may generate a selection signal SEL for controlling the sixth selector **409** to receive and output an input of the seventh pin **P47** of the memory device **40_7** based on the die selection signal CEBR and corresponding information.

(206) In response to receiving the read command within the third command **9005**, the memory device **40_7** may perform an internal operation for preparing data corresponding to the received third command **9005**. Accordingly, the memory device **407** may be in a busy state.

(207) At **t911**, the chip enable signal CEO may transition from a low level to a high level.

(208) At **t913**, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a fourth command **9007** to the memory package **200** through the data signal DQ. The fourth command **9007** may include an entire status read command **90h**. The package controller **35** may generate a selection signal SEL based on the command CMD and corresponding information. In this case, the selection signal SEL may be a signal that controls the sixth selector **409** so that signals inputted through the input pins of the sixth selector **409** corresponding to the first data pins (**P47a_0**, **P47a_1**, . . . , **P47a_n**) of the plurality of memory devices **40** are outputted through the seventh pins (**P27_0**, **P27_1**, . . . , **P27_7**).

(209) In some example embodiments, the memory device **400** may output status data indicating the

status of the memory device **40_0** at a time point of receiving the read status command **90h** (for example, **t914**).

(210) In some example embodiments, the plurality of memory devices **40** may output entire status data **9009** in response to receiving the fourth command **9007**. The entire status data **9009** may indicate the status of each of the memory devices (**40_0**, **40_1**, . . . , **40_7**) at **t914**. Specifically, because the memory device **400** is in a ready state, the seventh pin **P27_0** may output a bit (for example, 1) indicating the ready state. Because the memory devices (**40_1**, . . . , **40_7**) at **t914** are in a busy state, the seventh pin (**P27_1**, . . . , **P27_7**) may output a bit (for example, 0) indicating the busy state.

(211) The memory controller **10** may detect status data of each of the memory devices **40** based on the entire status data **9009** outputted from the memory package **200** in response to the fourth command **9007**.

(212) Subsequently, at **t915**, the chip enable signal CEO may transition from a low level to a high level.

(213) At **t917**, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a fifth command **9011** to the memory package **200** through the data signal DQ. The fifth command **9011** may include an entire status read command **90h**. The package controller **35** may generate a selection signal SEL based on the command CMD and corresponding information. For example, the selection signal SEL may be a signal that controls the sixth selector **409** so that signals inputted through the input pins of the sixth selector **409** corresponding to the first data pins (**P47a_0**, **P47a_1**, . . . , **P47a_n**) of the plurality of memory devices **40** are outputted through the seventh pins (**P27_0**, **P27_1**, . . . , **P27_7**).

(214) In some example embodiments, the plurality of memory devices **40** may output the entire status data **9009** in response to receiving the fifth command **9011**. The entire status data **9009** may indicate the status of the memory devices (**40_0**, **40_1**, . . . , **40_7**) at **t918**. Specifically, because the memory devices **40_0** and **40_1** are in a ready state, the seventh pins **P27_0** and **P27_1** may output a bit (for example, 1) indicating the ready state. Because the memory devices (**40_2**, . . . , **40_7**) at **t914** are in a busy state, the seventh pin (**P27_2**, . . . , **P27_7**) may output a bit (for example, 0) indicating the busy state.

(215) Then, at **t919**, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a sixth command **9015** to the memory package **200** through the data signal DQ. The sixth command **9015** may include a die selection signal CEBR for selecting die **0**. The package controller **35** may generate a selection signal SEL based on the die selection signal CEBR and corresponding information. For example, the selection signal SEL may be a signal that controls the sixth selector **409** so that signals inputted through the input pins of the sixth selector **409** corresponding to the first data pins (**P47a_0**, **P47a_1**, . . . , **P47a_n**) of the memory device **40_0** are outputted through the seventh pins (**P27_0**, **P27_1**, . . . , **P27_7**).

(216) Then, the memory device **40_0** may output data **9017** corresponding to the first command **9001** through the seventh pin **P47_0**. The interface circuit **32** may output data outputted from the memory device **40_0** as the data **9017** through the sixth selector **409** and the plurality of seventh pin **P27**.

(217) FIG. **10** illustrates a memory package according to an example embodiment.

(218) FIG. **10** illustrates a connection relationship between the interface circuit **33** and the plurality of memory devices **40** in a memory package **300**.

(219) In FIG. **10**, it is assumed that one memory package **300** includes eight memory devices (**40_0**, **40_1**, . . . , **40_7**). The present disclosure is not limited thereto, and one memory package **300** may include any number of memory devices.

(220) Each of the plurality of memory devices **40** may include first pins (**P41_0**, **P41_1**, . . . ,

P41_n; P41), a plurality of data pins (P47₀, P47₁, . . . , P47_n; P47), and eighth pins (P48₀, P48₁, . . . , P48_n; P48). FIG. 10 illustrates that one memory device, for example, the memory device 40₀ includes eight data pins (P47a₀, P47b₀, . . . , P47h₀), but the present disclosure is not limited thereto, and each of the plurality of memory devices 40 may include an arbitrary number of data pins.

(221) Hereinafter, descriptions of the memory device 40₀ may be equally applied to each of the memory devices 40.

(222) As shown in FIG. 10, the interface circuit 33 may include a plurality of second selectors 402 (402_a, 402_b, . . . , 402_h), a seventh selector 411, and an eighth selector 413.

(223) The interface circuit 33 may receive the chip enable signal CE from the memory controller 10 through the first pin P21. The first pin P21 may be connected to a plurality of first pins P41 of the plurality of memory devices 40.

(224) The interface circuit 33 may receive the data signal DQ from the memory controller 10 through a plurality of seventh pins P27. Alternatively, the interface circuit 33 may output data DATA from each of the memory devices 40_m through the plurality of seventh pins P27.

(225) The plurality of memory devices 40 may receive the chip enable signal CE and the plurality of data signals DQ from the interface circuit 33. The package controller 35 may generate a first selection signal SEL1 for controlling a plurality of second selectors 402.

(226) The seventh selector 411 may receive a ready/busy signal from the eighth pin P48 of the plurality of memory devices 40. The seventh selector 411 may generate RNB data RNB_DATA based on the received ready/busy signal. The seventh selector 411 may transmit the RNB data RNB_DATA to the package controller 35. The package controller 35 may generate a fourth signal C4 for controlling the eighth selector 413 based on the command CMD and the RNB data RNB_DATA. The fourth signal C4 may control the output received from the plurality of second selectors 402 to be output through the plurality of seventh pins P27, or may control the eighth selector 413 so that the eighth selector 413 may output data included in the fourth signal C4.

(227) The interface circuit 33 may output a ready/busy signal received from the plurality of memory devices 40 through the eighth pin P28.

(228) In some example embodiments, the plurality of memory devices 40 may receive a command CMD including a die selection command CEBR. For example, the memory controller 10 may transmit a status read command CMD including a die selection command CEBR indicating the memory device 40₀ to the memory device 40₀.

(229) The seventh selector 411 may be connected to a plurality of eighth pins P48 of each of the plurality of memory devices 40. The seventh selector 411 may output a value of "1" to the eighth pin P28 when one or more of the plurality of memory devices 40 is in a ready state (for example, when one or more of the outputs received from the plurality of eighth pins P48 has a value of "1"). In some example embodiments, the seventh selector 411 may be configured as an OR circuit.

(230) In some example embodiments, the plurality of memory devices 40 may receive a command CMD including a die selection command CEBR. For example, the memory controller 10 may transmit a read command CMD including a die selection command CEBR indicating the memory device 40₀ to the memory device 40₀.

(231) Each of the plurality of memory devices 40 may control the second selectors (402_a, 402_b, . . . , 402_h) and the eighth selector 413 so that the signals outputted from the plurality of memory devices 40 are outputted to the seventh pins (P27₀, P27₁, . . . , P27₇).

(232) The plurality of second selectors 402 may be connected to data pins of each of the plurality of memory devices 40. In addition, the plurality of second selectors 402 may be connected to each of the plurality of seventh pins (P27₀, P27₁, . . . , P27₇) of the interface circuit 33. In some example embodiments, each of the plurality of second selectors 402 may be configured as a multiplexer.

(233) The eighth selector 413 may be connected to the plurality of second selectors 402. The eighth

selector **413** may be connected to the plurality of seventh pins **P27**. In some example embodiments, the eighth selector **413** may connect the plurality of seventh pins **P27** to the output of the plurality of second selectors **402** based on the fourth signal **C4**. For example, the second selector **402_a** may be connected to the first data pin of each of the plurality of memory devices **40**. The second selector **402_a** may select one of the outputs received from the plurality of first data pins based on the first selection signal **SEL1** and output the selected signal to the eighth selector **413**.

Alternatively, the second selector **402_a** may output the data signal **DQ** received through the seventh pin **P27_0** to the first data pin of the memory device selected based on the first selection signal **SEL1**.

(234) As another example, the second selector **402_h** may be connected to the eighth data pin of each of the plurality of memory devices **40**. The second selector **402_h** may select one of the outputs received from the plurality of eighth data pins based on the first selection signal **SEL1** and output the selected signal to the eighth selector **413**. Alternatively, the second selector **402_h** may output the data signal **DQ** received through the seventh pin **P27_7** to the eighth data pin of the memory device selected based on the first selection signal **SEL1**.

(235) In another example embodiment, the eighth selector **413** may output data in the fourth signal **C4** to the plurality of seventh pins **P27** based on the fourth signal **C4**.

(236) For example, the package controller **35** may receive the ready/busy data **RNB_DATA** from the seventh selector **411** and generate the fourth signal **C4** based on the received ready/busy data **RNB_DATA**. The ready/busy data **RNB_DATA** may include status data of each of the plurality of memory devices **40**. A control logic **53** may detect information about memory devices in a ready state among the plurality of memory devices **40** based on the received ready/busy data **RNB_DATA**, and may generate the fourth signal **C4** indicating the number of the memory devices in the ready state and information about the memory devices. The control logic **53** may transmit the generated fourth signal **C4** to the eighth selector **413**. The eighth selector **413** may sequentially output the number of the memory devices in the ready state and the information indicating the corresponding memory devices based on the fourth signal **C4**.

(237) FIG. **11** illustrates a timing diagram of a chip enable signal, a ready/busy signal of a memory device, a total ready/busy signal, and a data signal according to FIG. **10**.

(238) Specifically, FIG. **11** illustrates a chip enable signal **CEO**, ready/busy signals (**RNB_0**, **RNB_1**, . . . , **RNB_7**) respectively outputted from a plurality of memory devices **40** in a memory package **20** selected by the chip enable signal **CEO**, a total ready/busy signal **RNB**, and a data signal **DQ** transmitted to the plurality of memory devices **40** through an interface circuit **30**.

Hereinafter, descriptions of the memory device **40_0** may be equally applied to each of the memory devices **40**.

(239) In some example embodiments, the ready/busy signal **RNB_0** may indicate the status of the memory device **40_0**. That is, the ready/busy signal **RNB_0** may be a signal outputted from the eighth pin **P48_0** of the memory device **40_0**. When the ready/busy signal **RNB_0** is at a low level, the memory device **40_0** may indicate a busy state, and when the ready/busy signal **RNB_0** is at a high level, the memory device **40_0** may indicate a ready state.

(240) The total ready/busy signal **RNB** may be determined by the package controller **35** based on each of the ready/busy signals (**RNB_0**, **RNB_1**, . . . , **RNB_7**).

(241) In addition, the memory controller (**10** in FIG. **1**) may transmit a read command together with the read enable signal (**nRE** in FIG. **2**), and in response to this, the memory interface **41_0** of the memory device **40_0** may transmit the data **DATA** to the memory controller **10** by aligning with the toggle timing of the data strobe signal **DQs** based on the read enable signal **nRE**.

(242) First, at **t1101**, the chip enable signal **CEO** transitions to an enable level (for example, a low level). The memory controller **10** may select the memory package **20** through the chip enable signal **CEO**.

(243) While the chip enable signal **CEO** is maintained at the low level, the memory controller **10**

may transmit the command CMD through the plurality of data signals DQ. The memory controller **10** may transmit a first command **11001** to the memory package **300** through the data signal DQ. In this case, the memory package **300** may be a memory package selected by the chip enable signal CEO. The first command **11001** may include a die selection signal CEBR and a read command for selecting die **0**. The package controller **35** may generate a first selection signal SEL1 and a fourth signal C4 for connecting the memory controller **10** and the memory device **40_0** based on the die selection signal CEBR. The first selection signal SEL1 may be a signal for connecting the plurality of second selectors **402** to the seventh pin P47_0 of the memory device **40_0**. The fourth signal C4 may be a signal for controlling the eighth selector **413** to connect the plurality of second selectors **402** and the seventh pin P27.

(244) In response to receiving the read command within the first command **11001**, the memory device **400** may perform an internal operation for preparing data corresponding to the received first command **11001**. That is, the memory device **40_0** may be in a busy state. Accordingly, the ready/busy signal RNB_0 indicating the status of the memory device **40_0** may transition to a low level indicating the busy state after a predetermined period after receiving the first command **11001**.

(245) At t1103, the chip enable signal CEO may transition from a low level to a high level.

(246) At t1105, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a second command **11003** to the memory package **300** through the data signal DQ. The second command **11003** may include a die selection signal CEBR and read command for selecting die **1**. The package controller **35** may generate a first selection signal SEL1 and a fourth signal C4 for connecting the memory controller **10** and the memory device **40_1** based on the die selection signal CEBR. The first selection signal SEL1 may be a signal for connecting the plurality of second selectors **402** to the seventh pin P47_0 of the memory device **40_1**. The fourth signal C4 may be a signal for controlling the eighth selector **413** to connect the plurality of second selectors **402** and the seventh pin P27.

(247) In response to receiving the read command within the second command **11003**, the memory device **40_1** may perform an internal operation for preparing data corresponding to the received second command **11003**. Accordingly, the ready/busy signal RNB_1 indicating the status of the memory device **401** may transition to a low level after a predetermined period after receiving the second command **11003**.

(248) At t1107, the chip enable signal CEO may transition from a low level to a high level.

(249) At t1109, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a third command **11005** to the memory package **300** through the data signal DQ. The third command **11005** may include a die selection signal CEBR and a read command for selecting die **7**. The package controller **35** may generate a first selection signal SEL1 and a fourth signal C4 for connecting the memory controller **10** and the memory device **40_7** based on the die selection signal CEBR. The first selection signal SEL1 may be a signal for connecting the plurality of second selectors **402** to the seventh pin P47_0 of the memory device **40_7**. The fourth signal C4 may be a signal for controlling the eighth selector **413** to connect the plurality of second selectors **402** and the seventh pin P27.

(250) In response to receiving the read command within the third command **11005**, the memory device **407** may perform an internal operation for preparing data corresponding to the received third command **11005**. Accordingly, the ready/busy signal RNB_7 indicating the status of the memory device **407** may transition to a low level after a predetermined period after receiving the third command **11005**.

(251) At t1111, the chip enable signal CEO may transition from a low level to a high level.

(252) At t1113, the chip enable signal CEO may transition from a high level to a low level. While

the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a fourth command **11007** to the memory package **300** through the data signal DQ. The fourth command **11007** may include an entire status read command **90h**. The seventh selector **411** may generate ready/busy signal data RNB_DATA based on the status data of each of the plurality of memory devices **40** outputted from the eighth pin P**48** of each of the plurality of memory devices **40**. The package controller **35** may generate the fourth signal C**4** based on the ready/busy signal data RNB_DATA. Specifically, the package controller **35** may detect that the memory device **40_0** of the plurality of memory devices **40** is in a ready state based on the ready/busy signal data RNB_DATA. Then, the package controller **35** may generate the fourth signal C**4** including information that the number of the memory devices in the ready state (READY COUNT) is 1 and corresponding memory device information (READY INFO) is the memory device **40_0**.

(253) Accordingly, the package controller **35** may control the eighth selector **413** so that the eighth selector **413** may output the number of the memory devices (READY COUNT) in the ready state in the fourth signal C**4** and the corresponding memory device information (READY INFO) through the seventh pin P**27**.

(254) In some example embodiments, the memory device **400** may output status data indicating the status of the memory device **40_0** at a time point of receiving the read status command **90h** (for example, t**1114**).

(255) In some example embodiments, the package controller **35** may generate the fourth control signal C**4** so that the eighth selector **413** may output ready data **11009** through the plurality of seventh pins P**27** in response to receiving the fourth command **11007**. The ready data **11009** may include information indicating the number of memory devices of the plurality of memory devices **400**, **40_1**, . . . , **40_7** in the ready state (READY COUNT) and the corresponding memory device (that is, the memory device **40_0**) (READY INFO).

(256) The memory controller **10** may detect status data of each of the memory devices **40** based on the ready data **11009** outputted from the memory package **100** in response to the fourth command **11007**.

(257) When receiving the entire status read command CMD, if any memory device among the plurality of memory devices **40** is not in the ready state, the eighth selector **413** may generate the fourth signal C**4** including information indicating that the number of the memory devices in the ready state (READY COUNT) is 0.

(258) When receiving the entire status read command CMD, if a plurality of memory devices among the plurality of memory devices **40** are in the ready state, the eighth selector **413** may generate the fourth signal C**4** indicating the number of the memory devices in the ready state (READY COUNT) and information about the corresponding memory devices. In some example embodiments, the information about the corresponding memory device may be listed according to a time during which the ready state is maintained. When the time required to prepare data is similar for each memory device, the information about the corresponding memory device may be listed according to the order in which read commands are received. For example, the memory controller may sequentially transmit the read command to the memory device **40_0** and the memory device **40_1**. In this case, the memory device **40_0** and the memory device **40_1** may be maintained at a busy state in response to the read command. Because the memory device **40_0** receives the read command earlier than the memory device **40_1**, the memory device **400** may be in a ready state earlier. Accordingly, in this case, the fourth signal C**4** may include information indicating the memory device **40_1** after information indicating the memory device **40_0**.

(259) At t**1115**, the chip enable signal CEO may transition from a low level to a high level.

(260) At t**1117**, the chip enable signal CEO may transition from a high level to a low level. While the chip enable signal CEO is maintained at the low level, the memory controller **10** may transmit a fifth command **11011** to the memory package **300** through the data signal DQ. The fifth command **11011** may include an entire status read command **90h**. The package controller **35** may generate the

fourth signal **C4** including information that the number of the memory devices (READY COUNT) in the ready state among the memory devices for which status information has not been checked is 1 and the corresponding memory device information (READY INFO) is the memory device **40_1**. (261) Accordingly, the package controller **35** may control the eighth selector **413** so that the number of the memory devices (READY COUNT) in the ready state in the fourth signal **C4** and the corresponding memory device information (READY INFO) may be outputted through the seventh pin **P27**.

(262) In some example embodiments, the package controller **35** may generate the fourth control signal **C4** so that the eighth selector **413** may output ready data **11013** through the plurality of seventh pins **P27** in response to receiving the fifth command **11011**. The ready data **11013** may include information about a memory device in a ready state among the plurality of memory devices (**40_0**, **40_1**, . . . , **40_7**). The ready data **11013** may indicate memory devices (**40_1**, **40_1**, . . . , **40_7**) whose status data has not been checked among the statuses of the memory devices (**40_0**, **40_1**, . . . , **40_7**) at **t1118**.

(263) The memory controller **10** may detect status data of each of the memory devices **40** based on the ready data **11013** outputted from the memory package **300** in response to the fifth command **11011**. The memory package **300** may output the ready data **11013** in response to receiving the fifth command **11011**. The ready data **11013** may indicate the status of the memory devices (**40_0**, **40_1**, . . . , **40_7**) at **t1118**. Specifically, because the memory devices **40_0** and **40_1** are in a ready state, the seventh pins **P27_0** and **P27_1** may output a bit (for example, 1) indicating the ready state. Because the memory devices (**40_2**, . . . , **40_7**) at **t1118** are in a busy state, the seventh pin (**P27_2**, . . . , **P27_7**) may output a bit (for example, 0) indicating the busy state.

(264) Then, at **t1119**, the chip enable signal **CEO** may transition from a high level to a low level. While the chip enable signal **CEO** is maintained at the low level, the memory controller **10** may transmit a sixth command **11015** to the memory package **300** through the data signal **DQ**. The sixth command **11015** may include a die selection signal **CEBR** for selecting die **0**. The package controller **35** may generate a first selection signal **SEL1** and a fourth signal **C4** for connecting the memory controller **10** and the memory device **40_0** based on the die selection signal **CEBR** and the ready/busy signal data **RNB_DATA**. For example, the first selection signal **SEL1** may be a signal for connecting the plurality of second selectors **402** to data pins of the memory device **40_0**. In addition, the fourth signal **C4** may be a signal for controlling the eighth selector **413** to connect the plurality of second selectors **402** to the seventh pin **P27**.

(265) Then, the memory device **40_0** may output data **11017** corresponding to the first command **11001** through the seventh pin **P47_0**. The interface circuit **33** may output the data outputted from the memory device **40_0** as the data **11017** through the eighth selector and the plurality of second selectors **402**.

(266) FIG. 12 illustrates a block diagram of an SSD system according to an example embodiment.

(267) Referring to FIG. 12, an SSD system **1000** may include a host **1100** and a SSD **1200**. The SSD **1200** transmits and receives signals to/from the host **1100** through a signal connector (e.g., interface), and receives a power source through a power connector (e.g., interface). The SSD **1200** may include an SSD controller **1210**, an auxiliary power supply **1230**, a buffer memory **1240**, and memory devices (**1221**, **1222**, . . . , **122n**). The memory devices (**1221**, **1222**, . . . , **122n**) may be NAND flash memory devices. In some example embodiments, the arbitrary memory devices (**1221**, **1222**, . . . , **122n**) may include the interface circuits according to FIG. 1 to FIG. 11.

(268) In this case, the SSD controller **1210** may transmit an entire status read command to all of a plurality of memory devices (**1221**, **1222**, . . . , **122n**) to detect the status of each of the plurality of memory devices (**1221**, **1222**, . . . , **122n**). In addition, the plurality of memory devices **1221**, **1222**, . . . , **122n** may each output status data in response to the entire status read command. The SSD controller **1210** may receive status data from the plurality of memory devices (**1221**, **1222**, . . . , **122n**). The SSD controller **1210** may output status data of each of the plurality of memory devices

1221, 1222, . . . , 122n as one entire status data to the host **1100** through a plurality of data pins.

(269) In related devices, the SSD controller may detect the status of one memory device at a time by transmitting a status read command together with one die selection signal. That is, the SSD controller requires n die selection signals and n status read commands in order to detect the statuses of all of the plurality of memory devices.

(270) However, because the SSD controller **1210** according to example embodiments of the present disclosure may detect the status data of all of the plurality of memory devices (**1221, 1222, . . . , 122n**) by using the entire status read command, fast data processing between the SSD controller **1210** and the plurality of memory devices (**1221, 1222, . . . , 122n**) is possible. In addition, the SSD controller **1210** according to the example embodiment of the present disclosure outputs the status data of the plurality of memory devices **1221, 1222, . . . , 122n** through existing data pins, thereby enabling fast data processing without adding new pins.

(271) In some example embodiments, each of the components represented by a block as illustrated in FIGS. **1-3, 5, 8, 10** and **12** may be implemented as various numbers of hardware, software and/or firmware structures that execute respective functions described above, according to example embodiments. For example, at least one of these components may include various hardware components including a digital circuit, a programmable or non-programmable logic device or array, an application specific integrated circuit (ASIC), transistors, capacitors, logic gates, or other circuitry using use a direct circuit structure, such as a memory, a processor, a logic circuit, a look-up table, etc., that may execute the respective functions through controls of one or more microprocessors or other control apparatuses. Also, at least one of these components may include a module, a program, or a part of code, which contains one or more executable instructions for performing specified logic functions, and executed by one or more microprocessors or other control apparatuses. Also, at least one of these components may further include or may be implemented by a processor such as a central processing unit (CPU) that performs the respective functions, a microprocessor, or the like. Functional aspects of example embodiments may be implemented in algorithms that execute on one or more processors. Furthermore, the components, elements, modules or units represented by a block or processing steps may employ any number of related art techniques for electronics configuration, signal processing and/or control, data processing and the like.

(272) While aspects of example embodiments have been particularly shown and described, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. A storage device comprising: a plurality of memory devices configured to output status data indicating a busy state indicating an internal operation is being performed or a ready state in response to a first command, and output error data indicating whether an error occurs in an internal configuration in response to a second command; and an interface circuit configured to receive the first command and the second command, receive the status data of each of the plurality of memory devices from the plurality of memory devices based on the first command, and output entire status data indicating statuses of the plurality of memory devices based on the status data of each of the plurality of memory devices, wherein n is an integer greater than 1, wherein the plurality of memory devices comprises n memory devices, wherein each of the plurality of memory devices comprises a status output pin through which the status data is output, and an error output pin, and wherein the interface circuit comprises: n output pins; and a first multiplexer configured to selectively connect the status output pin of each of the plurality of memory devices and the n output pins in response to the first command, and connect the error output pin of each of the plurality of memory devices and the n output pins in response to the second command.

2. The storage device of claim 1, wherein each of the plurality of memory devices is configured to, in response to receiving the second command, output error data through the error output pin, and wherein the interface circuit comprises a second selector that is connected between the first multiplexer and the n output pins, and is configured to selectively connect the error output pin of each of the plurality of memory devices and the n output pins in response to the second command.
3. The storage device of claim 1, wherein each of the plurality of memory devices comprises a plurality of data pins, and is configured to, in response to receiving a third command requesting data stored in the plurality of memory devices, output read data through the plurality of data pins, and wherein the interface circuit comprises a plurality of third selectors connected to the plurality of data pins and configured to be controlled by a second selection signal, and a fourth selector configured to selectively connect the plurality of third selectors and the plurality of data pins in response to a first control signal.
4. The storage device of claim 3, wherein the first control signal indicates a number of memory devices in the ready state among the plurality of memory devices and information identifying the memory devices.
5. The storage device of claim 4, wherein the first control signal indicates the number of memory devices in the ready state is 0 when no memory device is in the ready state among the plurality of memory devices.
6. The storage device of claim 4, wherein the first control signal, when more than one of the plurality of memory devices are in the ready state, comprises information identifying each of the memory devices in the ready state.
7. A storage device comprising: a memory controller configured to transmit a first command and a second command; and a memory package comprising: a plurality of memory devices configured to output status data in response to the first command, and output error data indicating whether an error occurs in an internal configuration in response to the second command; an interface circuit connected between the plurality of memory devices and the memory controller, and configured to receive the status data from the plurality of memory devices and output a plurality of status data to the memory controller; and a package controller configured to generate a selection signal to control the interface circuit, wherein the status data indicates whether a corresponding memory device is in a ready state or a busy state in which an internal operation is being performed, wherein n is an integer greater than 1, wherein the plurality of memory devices comprises n memory devices, and each of the plurality of memory devices comprises a status output pin and an error output pin, and wherein the interface circuit comprises n output pins and a first multiplexer configured to selectively connect the status output pin of each of the plurality of memory devices to the n output pins based on the selection signal and the first command, and connecting the error output pin of each of the plurality of memory devices and the n output pins based on the selection signal and the second command.
8. The storage device of claim 7, wherein each of the plurality of memory devices comprises a plurality of data pins, and is configured to, in response to receiving a third command, output read data through the plurality of data pins, wherein the interface circuit comprises a plurality of second selectors connected to n output pins and the plurality of data pins, and is configured to selectively connect the plurality of data pins to the n output pins based on a second selection signal, and wherein the package controller is further configured to generate the second selection signal in response to the third command.
9. The storage device of claim 8, wherein the interface circuit comprises the n output pins, and a third selector configured to receive the error data from each of the plurality of memory devices in response to the second command and output a plurality of error data through the n output pins based on the selection signal and the second command, and wherein the package controller is further configured to generate the selection signal in response to the second command.
10. The storage device of claim 8, wherein the interface circuit further comprises a fourth selector

connected between the plurality of second selectors and the n output pins, and configured to be controlled by a first control signal, and wherein the first control signal indicates a number of memory devices in the ready state among the plurality of memory devices and information identifying the memory devices.

11. The storage device of claim 10, wherein the fourth selector is further configured to connect the n output pins and the plurality of second selectors in response to the second command.

12. An operating method of a storage device, comprising: receiving, by each of a plurality of memory devices, a first command, from a memory controller; outputting, by each of the plurality of memory devices, status data indicating whether a corresponding memory device is in a ready state or a busy state in which an internal operation is being performed, through a status output pin in response to receiving the first command; generating a selection signal to control a plurality of selectors in an interface circuit to output the status data from each of the plurality of memory devices to the memory controller based on the first command; receiving, by each of a plurality of memory devices, a second command, from a memory controller; outputting, by each of the plurality of memory devices, error data through an error output pin in response to receiving the second command, wherein the interface circuit includes n output pins, wherein the plurality of selectors includes a plurality of status output pins, and a first multiplexer that is connected to the n output pins and is controlled by a first selection signal, and wherein the generating the selection signal comprises generating the first selection signal to control the first multiplexer to connect the plurality of status output pins and the n output pins in response to the first command.

13. The operating method of the storage device of claim 12, further comprising: receiving, from the memory controller, a second command requesting error data indicating whether an error occurs in an internal configuration of each of the plurality of memory devices.

14. The operating method of the storage device of claim 13, wherein the interface circuit includes a plurality of error output pins and the plurality of selectors includes a second selector that is connected to the n output pins and is controlled by a second selection signal, and wherein the generating the selection signal comprises generating the second selection signal to connect the plurality of error output pins and the n output pins in response to the second command.

15. The operating method of the storage device of claim 13, further comprising: receiving a third command requesting data in the plurality of memory devices from the memory controller; and outputting, by each of the plurality of memory devices, read data through a plurality of data pins in response to receiving the third command.

16. The operating method of the storage device of claim 15, wherein the plurality of selectors includes a plurality of third selectors connected to the plurality of data pins and controlled by a second selection signal, and a fourth selector connected to the plurality of third selectors and the n output pins and controlled by a first control signal, and wherein the generating the selection signal comprises generating the first control signal in response to the third command.

17. The operating method of the storage device of claim 16, wherein the generating the first control signal comprises inputting a number of memory devices in the ready state among the plurality of memory devices and information on the memory devices to the fourth selector in response to the first command.
